Commercial printing papers

By :

Raw materials

Grade A Paper: This is a high-quality paper known for it’s excellent smoothness and whiteness, often used in applications requiring superior print quality. The raw materials for **Grade A paper** typically consist of a blend of virgin wood fibres, sourced from both hardwood and softwood trees. The fibres are chosen for their length and strength, contributing to the final product durability and texture.

TMP Grade Paper: TMP refers to a method of producing wood pulp through the combination of heat and mechanical action. TMP grade paper is preferred for it’s high bulk, opacity, and ability to retain ink, making it suitable for books and other printed media that require both printability and durability without the high cost associated with fully chemical pulping processes.

Bond - process

Step 1: Pulping Process

For both types of paper, the first major step in manufacturing is pulping. For Grade A paper, this involves a chemical pulping process, typically using the kraft process, which effectively removes lignin and impurities from the wood fibres to ensure a cleaner, stronger paper. The kraft process is known for it’s efficiency and environmental friendliness due to it’s ability to recycle process chemicals.

In contrast, TMP grade paper uses a mechanical pulping process. Wood chips are heated with steam and then mechanically ground. The heat softens the lignin, and the mechanical action fibrillates the wood fibres, creating a pulp that retains most of it’s lignin, hence the higher yield and lower cost compared to chemical pulps.

Bond - process

Step 3: Paper Formation

Once the pulping is complete, the next step is forming the paper. The pulp is diluted with water and spread onto a wire mesh conveyor belt. As the belt moves, the water drains through the mesh, and the fibres begin to bond through natural hydrogen bonding, forming a wet mat of paper. During this phase, additives can be introduced to improve the paper's properties. For example, fillers like calcium carbonate are added to improve the brightness and opacity of the paper, while sizing agents are used to improve the paper's ink retention capabilities and moisture resistance.

Step 4: Pressing and Drying

After formation, the paper mat passes through a series of heavy rollers in the pressing section, which squeeze out excess water and begin to consolidate the sheet into a denser form. Following pressing, the paper enters the drying section, where it is subjected to heated rollers or cylinders that evaporate the remaining moisture. This part of the process is crucial for setting the paper's final properties.

Bond - process

Step 5: Calendaring and Winding

Post-drying, the paper is passed through calendar rollers. These heavy, smooth rollers compress and smooth the paper to achieve the desired thickness and surface texture. For Grade A paper, more intensive calendaring is performed to achieve a high degree of smoothness and uniformity essential for premium printing tasks.Finally, the continuous sheet of paper is slit according to the required widths and wound onto large reels to create giant rolls. These large rolls can be cut further into smaller sizes based on customer requirements.

Step 6: Quality Control and Packaging

The final steps involve rigorous quality, where the paper is tested for many properties like thickness, strength, smoothness, colour, and ink absorption. Only after meeting all the stringent criteria are the paper rolls packaged and prepared for shipping to distributors or end-users.

Bonded - advantages

- Bond paper can be environmentally friendly, especially if made from recycled components or if they come from sustainable certified sources. There are many brands that now provide recycled printing paper alternatives that decrease your carbon footprint while still offering you a good quality paper.

- Thanks to its composition, bond paper is more resilient to wear and tear compared to standard paper. It's resistant to ink bleeding and can withstand frequent handling.

- The smooth surface of bond paper ensures clear, crisp prints and vivid colors, making it perfect for high-quality documents and presentations.

- Bond paper works well with all types of printers, including inkjet and laser printers, making it a versatile choice for different printing needs.

Bonded - disadvantages

Physical and durability disadvantages

- Not durable: Without a protective coating, bond paper is easily scuffed, torn, or stained.
- Prone to damage: Lower-quality bond paper can curl or tear during printing.
- Not ideal for books: Heavier bond paper, when used for books, can make them very heavy and prone to damage, with pages that are easily dogeared or torn.

Cost and maintenance disadvantages

- Higher ongoing costs: Using ink or ribbons on bond paper adds to the running costs.
- Requires more maintenance: Bond paper printers require more maintenance compared to thermal printers.

Performance disadvantages

- Slower printing speeds: Printing on bond paper is generally slower, particularly with impact printers.
- Ink/toner issues: It requires regular ink changes.
- Potential for poor print quality: Using the wrong weight or low-quality paper can lead to issues like ink bleeding.

Bonded uses

- **Office Supplies:** Used extensively for letterheads, business correspondence, and reports where professional appearance is crucial.
- **Legal Documents:** Its durability makes it suitable for contracts and official forms where permanence is a requisite.
- **Creative Projects:** Due to its wide range of finishes and colors, it's a popular choice in crafting and scrapbooking circles for adding a touch of elegance.
- **Educational Materials:** Widely used in worksheets and exams due to its ability to withstand multiple handlings.

Coated - process

1. Preparing the coating mixture

- Components: Coating solutions are mixed in a "coating kitchen" according to precise recipes.
- Pigments: The main pigment is calcium carbonate or china clay, which gives the paper its white color. Talcum powder can also be used.
- Binders: These hold the pigments together and bind them to the paper. Examples include natural starches or synthetic compounds like latex.
- Additives: Other substances are added to give the paper specific properties, such as water resistance, which can include resins or polyethylene.

2. Coating the paper

- Application: The coating solution is applied to the paper as it travels through the paper machine. This can be a single, double, or triple coating process, depending on the desired smoothness.
- Blade coating: A common method is blade coating, where the coating is sprayed onto the paper and then scraped off with a steel blade. The pressure of the blade determines the uniformity of the surface.
- Drying: After each coating, the paper is dried, often using infrared heating panels and drying cylinders, to remove water.

Coated - process

Finishing the paper

- Calendering: The paper is then calendered, or milled, between hard rollers to increase its smoothness and create the final finish.
- Finishes: This calendering process can produce different finishes:
    - Gloss: Achieved with a high-gloss calendering process for a shiny appearance.
    - Matte: A less glossy, more subdued finish.
    - Silk: A finish that is a balance between gloss and matte.
- Winding and cutting: Finally, the paper is wound into large reels or cut into sheets, stacked, and prepared for shipping.

Coated - advantage

- Enhanced print quality: The coating fills the paper's natural valleys, creating a smoother surface that allows ink to sit on top instead of soaking in. This results in sharper, more vivid colors.
- Higher brightness: Pigments in the coating make the paper brighter and more opaque.
- Less ink usage: The improved ink holdout means less ink is needed to achieve the desired visual effect.
- Surface coating on coated paper can make it more resistant to dirt and moisture, but this depends on the “finish” of the coated paper. For example, gloss finish paper is more resistant to marking than matte or silk finishes.
- The only kind of coated paper that isn’t recyclable is paper coated in plastic. However, most coated paper used in commercial printing uses minerals. Plastic-coated paper is typically used for food, like disposable coffee cups. The same rules apply to composting.

Coated - disadvantage

- Poor writeability: The smooth, non-absorbent surface makes it difficult to write on with pens and pencils, which may smudge or fail to dry properly.
- Glare: Glossy coatings can create a high-gloss, reflective surface that makes it hard to read text, especially in areas with lots of words or small details.
- Cost: Coated paper generally costs more than uncoated paper because of the additional processing required to apply the coating.
- Environmental impact: The coating process can make the paper less environmentally friendly, and in some cases, its coatings can contaminate the recycling stream.
- Home printer incompatibility: Traditional coated papers are not suitable for most home inkjet or laser printers because the coatings can interfere with ink and toner adhesion.
- Limited folding and embossing: Heavily coated papers, especially cast-coated stock, can crack and are not ideal for projects that require folding.
- Not ideal for certain print techniques: It does not work well with some traditional print techniques like letterpress, which relies on the paper's ability to be impressed.

Coated - uses

- **Magazines and journals:** The smooth finish is ideal for printing high-resolution photographs and complex designs.
- **Brochures and flyers:** Coated paper, especially with a glossy finish, makes promotional materials stand out and look more professional.
- **Catalogs:** It is used for catalogs to ensure that detailed product images appear crisp and clear.
- **Photographs:** Its surface is perfect for photographic prints due to its ability to reproduce sharp, vibrant images.
- **Packaging:** The paper's durability and resistance to moisture and chemicals make it a good choice for packaging, especially in more demanding environments.
- **Book covers:** Coated paper provides a durable, high-quality look for book covers, which often feature heavy ink coverage and special finishes.
- **Marketing collateral:** Materials like branded inserts and other premium branding materials benefit from the polished look of coated paper.